

RE040 – Estimated Deployment and Operational Costs for Food Remedy API

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Introduction –

There is more to deploying a mobile application than just uploading it to the Play Store or App Store. For applications to function properly after release, they also need backend hosting, cloud storage, analytics systems, monitoring tools, and continuous maintenance. Planning for scalability and sustainability requires an awareness of the financial costs related to deployment and long-term operation as the Food Remedy API gets ready for production deployment.

The anticipated expenses for launching and maintaining the Food Remedy app on the Google Play Store and Apple App Store are examined in this study. Additionally, it looks at third-party service prices, backend infrastructure costs, and ways to save operating costs without sacrificing dependable application performance.

App Store Publishing Costs-

Developers must create official developer accounts with Google and Apple before an application may be made available to the public.

Google Play Store requires a one-time developer registration fee of approximately USD \$25, which is around AUD \$40. Developers do not have to pay annual fees to publish several applications under the same account once they have registered.

Developers must sign up for the Apple Developer Program, which currently costs USD \$99 per year, in order to use the Apple App Store. Maintaining publishing access and distributing apps via the App Store ecosystem requires this annual subscription charge.

Although these publishing costs are relatively affordable, they represent only the initial stage of deployment expenses.

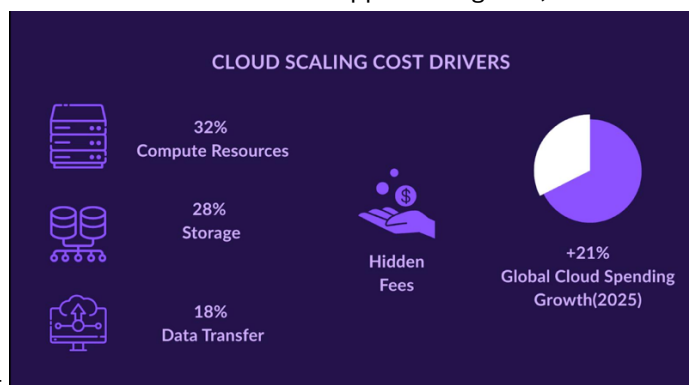
Backend Hosting and Infrastructure Costs

After deployment, the Food Remedy API will require backend infrastructure to support barcode scanning, recommendation generation, authentication services, analytics tracking, and database management. Cloud

platforms such as Firebase, Google Cloud Platform, AWS, and Render are commonly used because they provide scalable infrastructure and flexible pricing models suitable for mobile applications.

At first, low user traffic and early deployment may be supported by free-tier services. However, as the service gets more users and manages more traffic, operating costs progressively rise. Rising infrastructure costs over time are caused by a number of factors, including increased API calls, greater database storage needs, cloud processing, analytics tracking, and monitoring systems. Larger user bases necessitate more sophisticated server scalability, stronger backend systems, and more storage capacity, all of which raise monthly operating costs dramatically.

The Food Remedy API will also need cloud security measures, backup services, and monitoring systems to ensure consistent application performance. Long-term operational planning is required to guarantee that infrastructure expenses remain reasonable as the application grows, even when cloud architecture enables



scalable deployment.

Third-Party Service Costs

In order to enhance performance, monitoring, communication, and overall user experience, modern mobile applications frequently rely on third-party services. For recommendation-related functionality, the Food Remedy API may rely on services like Firebase Analytics, Crashlytics, email systems, push notification services, authentication systems, and AI-based APIs.

In the first phases of deployment, several of these services offer free-tier access. However, recurrent operating costs start to rise as user traffic rises and use beyond free restrictions. While cloud storage, authentication services, email APIs, and AI-powered systems can become increasingly costly based on request volume and processing needs, analytics solutions may initially remain inexpensive.

Third-party services raise long-term infrastructure dependency and operating costs even while they enhance scalability, dependability, and application quality. Therefore, sustaining a financially sustainable deployment environment requires careful monitoring of usage and the selection of cost-effective services.

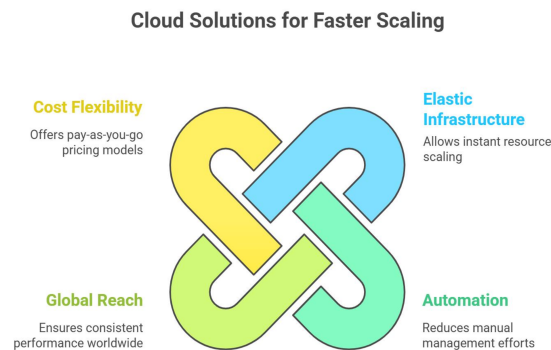
Cost Optimization Strategies

There are a number of ways to lower the Food Remedy API's deployment and operating costs without sacrificing dependable application performance. One of the most effective approaches is using scalable cloud infrastructure that allows the system to grow gradually instead of requiring large upfront investments. Platforms such as Firebase and Google Cloud provide flexible pricing models that support early-stage

deployment while allowing backend resources to scale according to user demand.

Another significant factor in lowering operating expenses is backend optimization. Caching technologies, compressed assets, efficient database searches, and fewer pointless API requests can all greatly cut server load and cloud processing costs. Regularly keeping an eye on infrastructure consumption also makes it easier to spot areas where resources are being squandered or exploited.

Gradual scaling is an additional crucial tactic. Instead of making significant initial investments in enterprise-level systems, the Food Remedy API can start with inexpensive infrastructure and gradually add more resources as the user base grows. With this strategy, the application may continue to be commercially viable while maintaining long-term production readiness and future scalability.



Conclusion-

Both the original publication costs and continuing operating costs following release are involved in the Food Remedy application's deployment. Backend infrastructure, cloud hosting, analytics systems, and third-party services add recurrent monthly costs that rise as the application grows, even though App Store and Play Store registration fees are reasonably priced.

This study showed that free-tier services and scalable cloud infrastructure can enable economical early implementation while permitting future expansion. Maintaining a financially sustainable production environment requires careful infrastructure planning, backend optimization, and gradual scaling techniques. The Food Remedy API can support long-term scalability and operational growth while maintaining dependable performance through the use of economical deployment techniques.

References-

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